

SLE-3: Architectural Design (Full C4 Model)

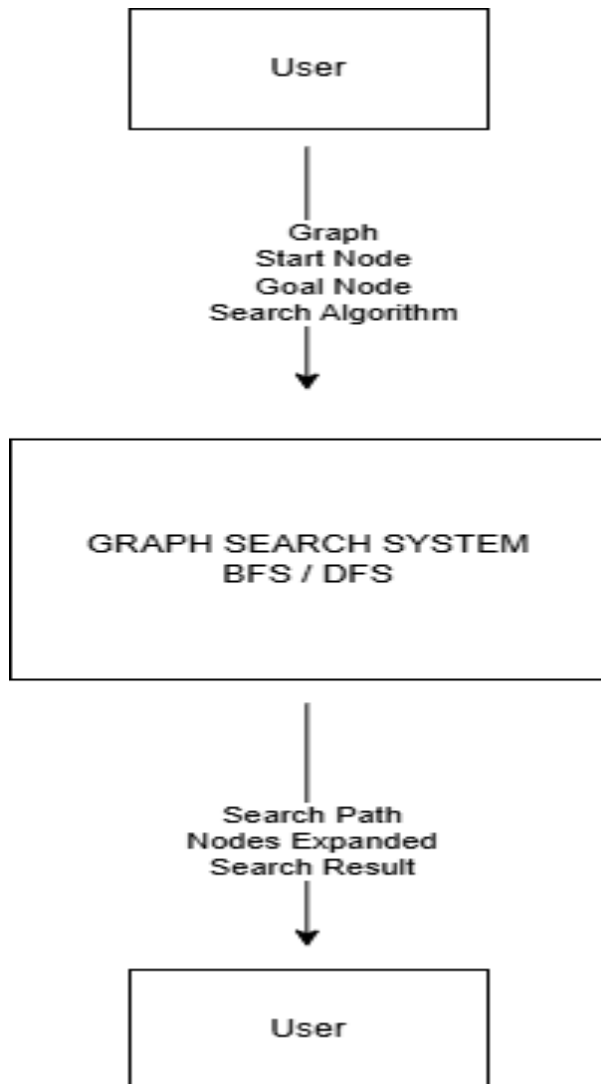
Course: 02AML204 – Introduction to Artificial Intelligence

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1. System Title & Short Description

Graph Search System using BFS and DFS is a search-based system developed to find a path between a start node and a goal node in a graph. The system supports Breadth-First Search (BFS) and Depth-First Search (DFS). It records the search path and number of nodes expanded. The project also measures execution performance and uses py-spy profiling for performance analysis.

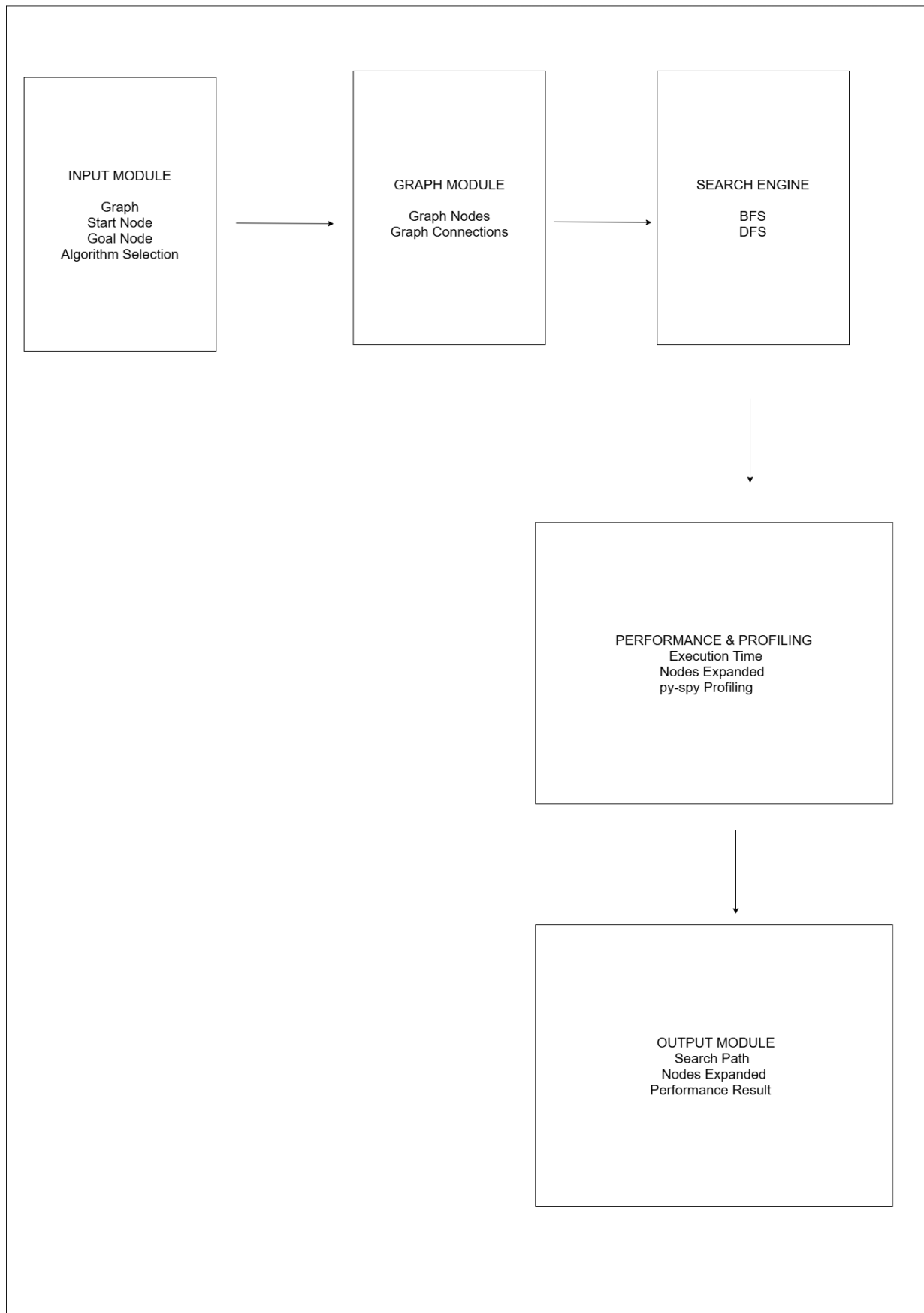
2. Context Diagram (Level 1)



LEVEL 1 DIAGRAM

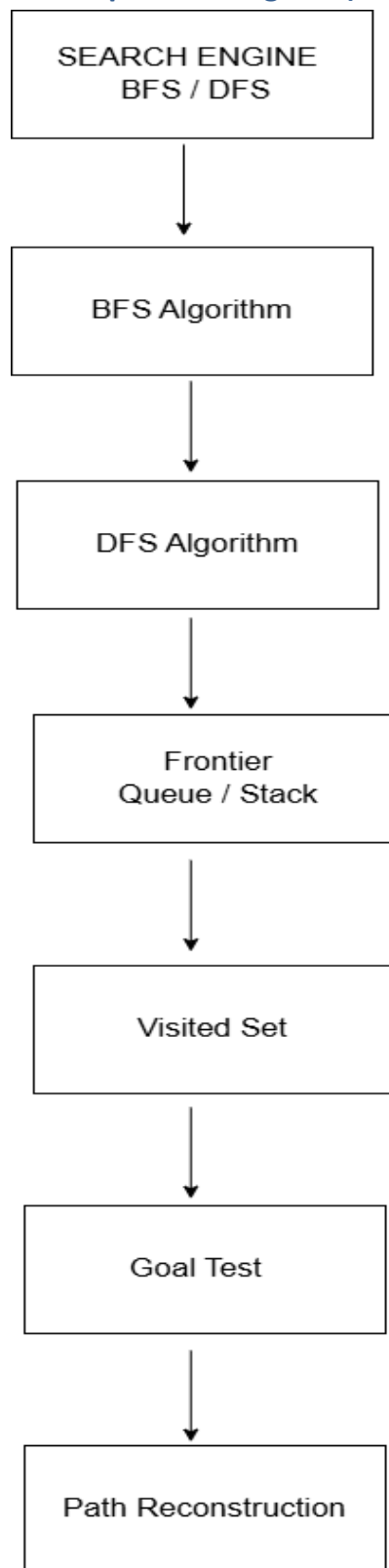
The user provides the graph, start node, goal node and selected search algorithm. The Graph Search System performs BFS or DFS and returns the search path, nodes expanded and search result.

3. Container Diagram (Level 2)



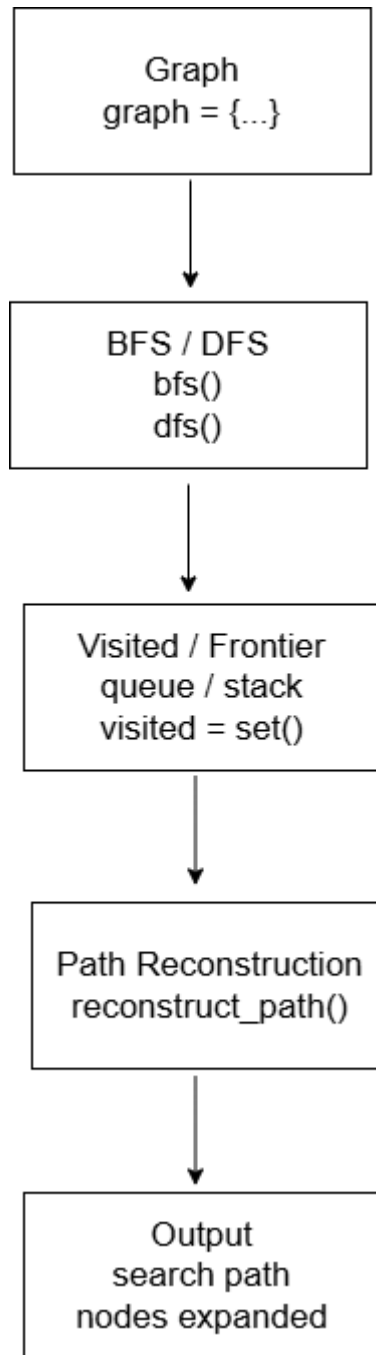
- Input Module: Accepts the graph, start node, goal node and selected algorithm.
- Graph Module: Stores the graph nodes and their connections.
- Search Engine: Executes BFS and DFS search algorithms.
- Performance & Profiling Module: Measures execution performance, nodes expanded and py-spy profiling.
- Output Module: Displays the search path, nodes expanded and performance result.

4. Component Diagram (Level 3)



The Search Engine is divided into components that perform the main search operations. The BFS and DFS components implement the two search strategies. The Frontier manages nodes waiting to be explored, while the Visited Set keeps track of explored nodes. The Goal Test checks whether the target has been reached, and Path Reconstruction produces the final path.

5. Code Level Overview (Level 4)



- graph = {...} : Stores graph nodes and connections.
- bfs() : Performs Breadth-First Search.
- dfs() : Performs Depth-First Search.
- queue / stack : Manages the frontier during search.
- visited = set() : Keeps track of explored nodes.
- reconstruct_path() : Generates the final search path.

6. Design Decisions

The system is divided into separate modules so that input, graph handling, search, performance measurement and output are easy to understand. BFS and DFS are kept inside the Search Engine because they are the main search strategies. A separate performance and profiling module supports the SLE-2 performance analysis.

7. AI Contribution Note

- AI tools used: ChatGPT
- What AI helped with: Understanding the C4 architecture, organizing the report structure and improving explanations.
- What I did myself: Implemented and tested the BFS/DFS graph search system, performed profiling, created the diagrams and verified the project results.

8. Conclusion

This SLE helped me understand how the same Graph Search System can be represented at four different C4 levels. I learned to describe the system from the overall user interaction to its containers, components and main code functions. The architecture also connects directly with my previous BFS/DFS implementation and performance analysis.